

A Simplified FSO Channel Model with Weak Turbulence and Pointing Errors

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ABSTRACT Free-space optical (FSO) links suffer from atmospheric turbulence and pointing errors. The probability density function (PDF) describing the distribution of the fading coefficient that models such effects is of great importance for designing FSO systems. In this paper, we present a simple closed-form approximation for this PDF in the weak turbulence regime. Our proposed approximation involves only exponential functions. This approximation is then used to develop two simple bit error rate expressions for on–off keying. Numerical results show excellent accuracy of the proposed expressions.

Keywords: Free-space optical (FSO), atmospheric turbulence, pointing errors, channel modeling.

1. INTRODUCTION

5G and the evolution towards 6G are expected to deliver mobile broadband services and connect billions of devices. To enable these services, the application of free-space optical (FSO) communications has recently gained great interest. This growing interest is thanks to the large available bandwidth, which results in ultra-high data rates [1]. Furthermore, FSO communications provide high security, no spectrum licensing, and low cost [2]. However, FSO links are subject to atmospheric turbulence, pointing errors [3], and atmospheric losses, which must be effectively accounted for to achieve their promised data rates. The channel model studied in this work considers these three effects as the main phenomena that independently create multiplicative losses that deteriorate the performance of the link.

Atmospheric turbulence is the result of small changes in air pressure and temperature, causing random variations of the atmospheric refractive index. These variations cause fluctuations in both phase and intensity of the received optical signal [4]. Several statistical models have been proposed in the literature to characterize this turbulence fading. These mathematical models can be classified according to their regions of validity. For weak turbulence, the intensity fluctuations are typically modelled by the log-normal distribution, while for strong turbulence, the K-distribution and the gamma-gamma distribution are widely used [5]. In this paper, we focus on the weak turbulence regime.

The second effect we consider in this paper is pointing errors. These errors are caused by the misalignment of the transmit beam with respect to the field of view of the receiver. This misalignment is due to small mechanical variations and meteorological phenomena such as strong winds, and it causes random movements in the relative position of transmitter and receiver. It is expected that future FSO systems will ensure small pointing errors as discussed in [6, Table 1], which is why this work considers low pointing errors. Finally, atmospheric loss is caused by the presence of particles that either absorb or scatter the transmitted light.

In the weak turbulence regime, the joint probability density function (PDF) that characterises the complete channel response determined by atmospheric turbulence, pointing errors, and atmospheric losses was developed in [3, Sec. III-D]. This PDF is based on the complementary error function. When this PDF is evaluated using small pointing errors and off-the-shelf numerical routines, numerical instabilities appear. Furthermore, the complementary error function prevents us from finding a simple expression for performance metrics such as the bit error rate (BER).

In this paper, we develop a simple closed-form approximation for the PDF proposed in [3]. Our expression is based on exponential functions only, and thus, it effectively solves the numerical problems inherent to the expression in [3] for small pointing errors. The second contribution of this paper is to use our approximated PDF to obtain an analytical BER expression for on–off keying (OOK) signaling. Finally, we show that our simplified channel model and BER expressions are in good agreement with the Monte-Carlo simulations results realized using the true PDF.

2. SYSTEM MODEL

We consider an FSO communication system employing intensity modulation (IM) with direct detection (DD) as illustrated in Fig. 1. IM is realized by first converting the identically and uniformly distributed information bits b_k at time instant k into OOK symbols x_k , where scaling by $2P$ results in $x_k \in \{0, 2P\}$. Here, P is the average transmit optical power. The OOK symbols are then interleaved by Π resulting in the symbol sequence to be transmitted. Within the electrical-to-optical (E/O) conversion block, a modulator generates an electrical waveform to directly modulate the light source. The optical signal is then transmitted through the FSO channel. At the receiver side, a photodetector (PD) detects the power of the signal. The received photocurrent is directly proportional to the incident optical power via the detector responsivity η . The demodulator outputs the sequence

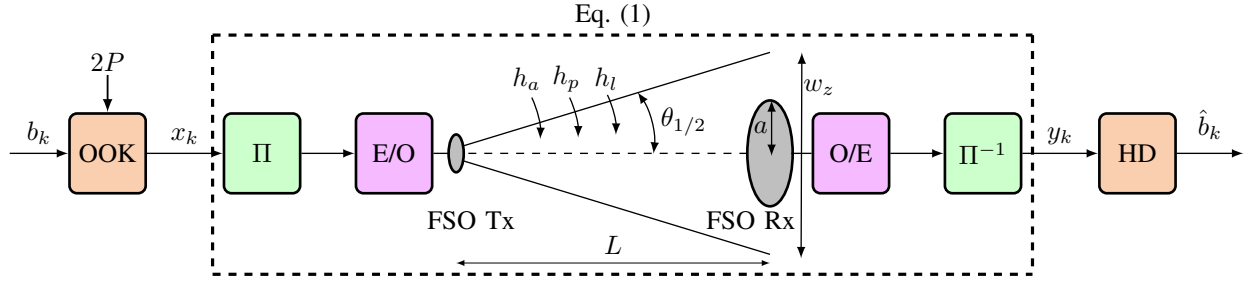


Figure 1: Block diagram of the IM/DD FSO communication system with OOK modulation under consideration. The channel effects consist of atmospheric turbulence (h_a), geometric spread and pointing errors (h_p), and atmospheric losses (h_l).

of symbols. Finally, after deinterleaving by Π^{-1} , the bit estimates $\hat{b}_k$ are obtained by making hard decisions (HD) on the symbols y_k .

Since the timescale of the fading processes (on the order of milliseconds [3, Sec. 2]) is far larger than the symbol duration (on the order of nanoseconds), the channel gain coefficient is constant over a large number of transmitted bits, effectively resulting in a slow fading channel. However, in this paper we assume an infinitely long interleaver, and thus, the input-output relationship between symbols x_k and y_k can be modelled as [3, eq. (1)]

$$y_k = \eta h_k x_k + n_k, \quad (1)$$

where n_k is a zero-mean signal-independent additive Gaussian noise with variance σ_n^2 .

The channel model in (1) is memoryless, and thus, from now on we remove the index k . The channel gain coefficient h in (1) models the random attenuation of the propagation channel. In our model, this attenuation is due to three factors: atmospheric turbulence h_a , geometric spread and pointing errors h_p , and atmospheric loss h_l . Since these phenomena create independent multiplicative losses, the channel gain coefficient can be formulated as [3, eq. (2)]

$$h = h_a h_p h_l. \quad (2)$$

Note that h_l in (2) is determined by the exponential Beers–Lambert Law [3], and is assumed to be constant. On the other hand, h_p and h_a are random variables (RVs) whose distributions will be discussed in the next section.

3. CHANNEL MODEL

3.1 Atmospheric Turbulence

Numerous studies have been carried out with the aim of developing mathematical models to characterize the PDF that can model the turbulence in a wide range of atmospheric conditions [5], [7]. For weak turbulence, the PDF is accepted to follow a log-normal distribution. The logarithm of the amplitude (often called log-amplitude in the literature) of the optical wave affected by weak turbulence has a Gaussian PDF [8, Sec. 8.1] with log-amplitude variance σ_X^2 given by [3, Sec. III-A]

$$\sigma_X^2 \approx \frac{\sigma_R^2}{4}, \quad (3)$$

where σ_R^2 is the Rytov variance defined as

$$\sigma_R^2 = 1.23 C_n(z)^2 k^{7/6} L^{11/6}. \quad (4)$$

In (4), L is the propagation distance, $C_n(z)^2$ the index of refraction structure parameter at altitude z from ground level (assumed to be constant along the propagation path for horizontal links), and $k = 2\pi/\lambda$ the optical wave number. The atmospheric turbulence PDF is given by [3, eq. (5)] and [4, eq. (3)] as

$$f_{H_a}(h_a) = \frac{1}{2h_a \sqrt{2\pi\sigma_X^2}} \exp\left(-\frac{(\ln(h_a) + 2\sigma_X^2)^2}{8\sigma_X^2}\right). \quad (5)$$

3.2 Pointing Errors and Geometric Spread

In line-of-sight FSO communication links, geometric spread and pointing accuracy also affect the performance of the system. The geometric spread is caused by the divergence of the transmit beam when propagating through the atmosphere as $w_z \approx \theta_{1/2} L$, with w_z being the received beam waist radius and $\theta_{1/2}$ denoting half of the transmitter divergence angle, see Fig. 1. Pointing worsens due to imperfections in the beam pointing, acquisition, and tracking processes.

Geometric spread and pointing accuracy are interrelated so that they are usually taken in to account jointly in many modelling frameworks. Here, we consider the model proposed in [3] which assumes a Gaussian beam with beam waist ω_z on the receiver plane and a circular aperture receiver of radius a , see Fig. 1. Then, the attenuation due to the geometric spread with pointing errors can be approximated as [3, eq. (9)]

$$H_p \approx A_0 \exp\left(-\frac{2R^2}{\omega_{z_{eq}}^2}\right), \quad (6)$$

where R is the radial displacement, A_0 is the fraction of the collected power without pointing errors, i.e., only due to geometric spread, and $\omega_{z_{eq}}^2$ is the equivalent beam width. Here, A_0 and $\omega_{z_{eq}}^2$ are given by $A_0 = \text{erf}(v)^2$ and $\omega_{z_{eq}}^2 = \omega_z^2 \sqrt{\pi} \text{erf}(v) / (2v \exp(-v^2))$, $\text{erf}(\cdot)$ is the error function, and $v = \sqrt{\pi}a / (\sqrt{2}\omega_z)$.

Moreover, considering independent identical Gaussian distributions for the horizontal x and y displacements in the receiver plane, the radial error $r = \sqrt{x^2 + y^2}$ is modeled via a Rayleigh distribution determined by σ_s^2 as the jitter variance at the receiver. Under these assumptions, the channel attenuation H_p can be seen as a function of the radial displacement R and has the PDF [3, eq. (11)]

$$f_{H_p}(h_p) = \frac{\gamma^2}{A_0^{\gamma^2}} h_p^{\gamma^2-1}, \quad 0 \leq h_p \leq A_0, \quad (7)$$

where

$$\gamma \triangleq \omega_{z_{eq}} / (2\sigma_s) \quad (8)$$

denotes the ratio between the equivalent beam radius at the receiver and the pointing error displacement standard deviation. This last parameter is estimated as $\sigma_s \approx \theta_s L$, where θ_s is the jitter angle.

3.3 Atmospheric Attenuation

The atmospheric loss can be computed by the well-known Beer-Lambert law given in [3, Sec. III-B] and [9, eq. (1)] as

$$h_l(L) = \exp(-\sigma(\lambda)L), \quad (9)$$

where $\sigma(\lambda)$ is a wavelength-dependent attenuation coefficient [9]. This coefficient depends on weather conditions and can be obtained from the atmospheric visibility according to Kim's model [10, eq. (3)]. Since the attenuation coefficient slowly changes over time, in our analysis, we have assumed that h_l is a deterministic coefficient.

3.4 Composite Channel Model

The distribution of $H = h_l H_p H_a$, where $f_{H_a}(h_a)$ and $f_{H_p}(h_p)$ are given by (5) and (7), resp., can be expressed as [3, eq. (14)]

$$f_H(h) = \frac{\gamma^2}{2(A_0 h_l)^{\gamma^2}} h^{\gamma^2-1} \text{erfc}\left(\frac{\ln(\frac{h}{A_0 h_l}) + \mu}{\sqrt{8\sigma_X^2}}\right) \exp[2\sigma_X^2 \gamma^2 (1 + \gamma^2)], \quad (10)$$

where

$$\mu \triangleq 2\sigma_X^2 (1 + 2\gamma^2), \quad (11)$$

and

$$\text{erfc}(z) \triangleq \frac{2}{\sqrt{\pi}} \int_z^\infty \exp(-t^2) dt, \quad (12)$$

defined as the complementary error function. In the next section, we study approximations for the PDF in (10).

4. PROPOSED APPROXIMATIONS

The PDF in (10) includes the complementary error function whose integral form is given in (12). This integral has no analytical solution and is often evaluated numerically, which prevents us from using (10) to obtain closed-form expressions of performance metrics such as BER. Moreover, when the PDF in (10) is evaluated numerically for small pointing errors (i.e., for small values of σ_s) using off-the-shelf numerical routines, numerical instabilities occur. More specifically, due to finite numerical precision, the computation of (10) may lead to the multiplication of zero and infinity during the routine, resulting in indeterminate values. In this section, we propose a simple closed-form approximation for the PDF in (10) that (i) avoids numerically instabilities, and (ii) leads to simple BER expressions.

We first apply the following approximation to the complementary error function

$$\text{erfc}(x) \approx \frac{1}{x} \frac{1}{\sqrt{\pi}} \exp(-x^2) \quad (13)$$

TABLE I: FSO System Configuration

FSO link		Transmitter and receiver	
Parameter	Value	Parameter	Value
Wavelength (λ)	1550 nm	Responsivity (η)	0.5
Tx-Rx Link distance (L)	3 km	Tx divergence half angle ($\theta_{1/2}$)	0.66 mrad
Noise standard deviation (σ_n)	10^{-7} A/Hz	Jitter angle (θ_s)	0.04 – 0.116 mrad
Attenuation coefficient ($\sigma(\lambda)$)	0.2208 dB/km	Receiver radius (a)	5 cm

introduced in [11, eq. (6)]. Inserting (13) into (10), $f_H(h)$ can be approximated as

$$f_H(h) \approx \begin{cases} \frac{\gamma^2 h^{\gamma^2-1}}{\sqrt{\pi}(A_0 h_l) \gamma^2} \frac{\sqrt{2\sigma_X^2}}{\ln(h/(h_l A_0)) + \mu} \exp \left[- \left(\frac{\ln(\frac{h}{h_l A_0}) + \mu}{\sqrt{8\sigma_X^2}} \right)^2 \right] \exp [2\sigma_X^2 \gamma^2 (1 + \gamma^2)], & \text{if } h > \hat{h}, \\ 0, & \text{otherwise,} \end{cases} \quad (14)$$

where $\hat{h} \triangleq A_0 h_l \exp(-\mu)$. We constrain the support of this approximation to $h > \hat{h}$ since outside this region, the approximation in (13) and hence the approximation PDF in (14) results in negative values. Consequently, the area underneath the approximate PDF in (14) is not equal to 1 in general. This threshold $\hat{h}$ is obtained by equating to zero the argument of the complementary error function in (10), i.e., by solving $(\ln(h/(h_l A_0)) + \mu)/\sqrt{8\sigma_X^2} = 0$.

We will now use (14) to derive simple BER expressions for the FSO system given in Fig. 1. We consider maximum likelihood detection with perfect channel state information, i.e., both h and η in (1) are assumed to be known at the receiver. We define the BER conditioned on $H = h$ as $P_b(h) \triangleq \Pr\{\hat{B} \neq B | H = h\}$, which can be expressed as

$$P_b(h) = \frac{1}{2} \Pr\{\hat{B} = 1 | B = 0, H = h\} + \frac{1}{2} \Pr\{\hat{B} = 0 | B = 1, H = h\} = \frac{1}{2} \operatorname{erfc} \left(\frac{\eta P h}{\sqrt{2\sigma_n^2}} \right). \quad (15)$$

The *exact* average BER $\bar{P}_b$ can be computed by integrating $P_b(h)$ in (15) over the exact PDF in (10), which yields

$$\begin{aligned} \bar{P}_b &= \int_0^\infty P_b(h) f_H(h) dh \\ &= \frac{\gamma^2}{4(A_0 h_l) \gamma^2} \exp [2\sigma_X^2 \gamma^2 (1 + \gamma^2)] \int_0^\infty h^{\gamma^2-1} \operatorname{erfc} \left(\frac{\eta P h}{\sqrt{2\sigma_n^2}} \right) \operatorname{erfc} \left(\frac{\ln(\frac{h}{A_0 h_l}) + \mu}{\sqrt{8\sigma_X^2}} \right) dh. \end{aligned} \quad (16)$$

An *approximate* average BER can be obtained by using the approximate PDF in (14), which yields

$$\bar{P}_b \approx \frac{\gamma^2}{(A_0 h_l) \gamma^2} \frac{\sigma_X}{\sqrt{2\pi}} \exp [2\sigma_X^2 \gamma^2 (1 + \gamma^2)] \int_{\hat{h}}^\infty \frac{h^{\gamma^2-1}}{\ln(\frac{h}{A_0 h_l}) + \mu} \operatorname{erfc} \left(\frac{\eta P h}{\sqrt{2\sigma_n^2}} \right) \exp \left[- \left(\frac{\ln(\frac{h}{A_0 h_l}) + \mu}{\sqrt{8\sigma_X^2}} \right)^2 \right] dh. \quad (17)$$

We then again apply the approximation to erfc given in (13) to (17) to further simplify it, which yields

$$\bar{P}_b \approx \frac{\gamma^2 \sigma_X \sigma_n}{\eta P \pi (A_0 h_l) \gamma^2} \exp [2\sigma_X^2 \gamma^2 (1 + \gamma^2)] \int_{\hat{h}}^\infty \frac{h^{\gamma^2-2}}{\ln(\frac{h}{A_0 h_l}) + \mu} \exp \left(- \frac{(\eta P h)^2}{2\sigma_n^2} - \left(\frac{\ln(\frac{h}{A_0 h_l}) + \mu}{\sqrt{8\sigma_X^2}} \right)^2 \right) dh. \quad (18)$$

Similar to (14), our proposed average BER approximations in (17) and (18) are free of numerical instabilities for the entire pointing errors regime.

5. NUMERICAL RESULTS

In this section, the average BER expressions in (16), (17), and (18) are compared. The aim of this analysis is to evaluate the accuracy of the proposed approximations under realistic conditions. Table I summarizes the parameters considered in our analysis. These parameters are obtained from [3], [6], [12].

First, in Fig. 2 (left), we identify the regions of pointing errors and turbulence in which (10) may lead to numerical instabilities. On the x-axis, the severity of pointing errors is varied from $\sigma_s = 0.12$ m to 0.35 m corresponding to jitter angles of $\theta_s = 0.04$ mrad and $\theta_s = 0.116$ mrad, respectively. This left extreme is labeled *deluxe* system in [6, Table 1]. On the y-axis, the severity of turbulence is constrained to $\sigma_R^2 \leq 1$ which corresponds to the weak turbulence regime [8, Fig. 7.4]. Two regions are distinguished: The red area (left-hand side) corresponds to the region associated with very low pointing errors where numerical instabilities appear when off-the-shelf numerical routines are used to evaluate (10). For the blue area (right-hand side), these instabilities disappear.

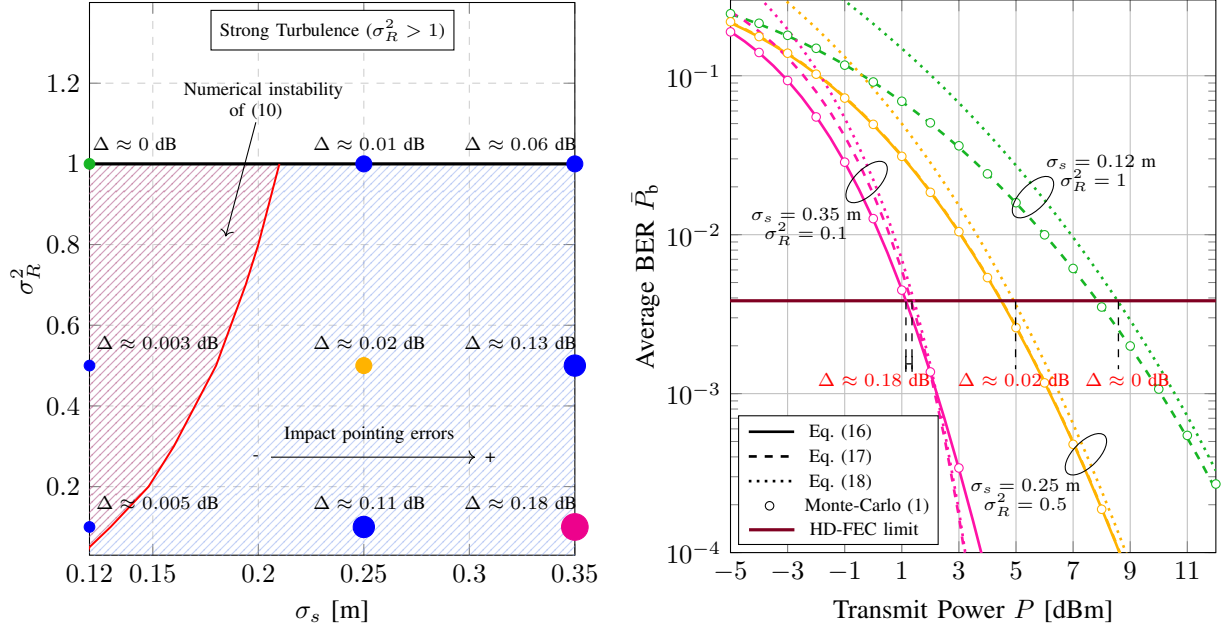


Figure 2: (Left) Region of application for a 3 km link considering weak turbulence regime and $\theta_s = 0.04$ mrad and 0.116 mrad that determine the horizontal axis as $\sigma_s \approx \theta_s L$. (Right) Average BER versus transmitted power considering clear air, $L = 3$ km, and three different sets of (σ_s, σ_R^2) : (0.12, 1), (0.25, 0.5), and (0.35, 0.1).

Then, in Fig. 2 (right), the average BERs $\bar{P}_b$ in (16), (17), and (18) are plotted as a function of the transmitted power P . In terms of pointing errors and turbulence, three operating points are considered. For the first, $\sigma_s = 0.12$ and $\sigma_R^2 = 1$ with corresponding point and curves colored with green in Fig. 2 (left-right), respectively. For the second, $\sigma_s = 0.25$ and $\sigma_R^2 = 0.5$ with yellow color. For the third, $\sigma_s = 0.35$ and $\sigma_R^2 = 0.1$ with pink color.

In Fig. 2 (right), solid, dashed, and dotted curves correspond to the exact average BER (16) and approximate average BERs (17) and (18), respectively. BER estimates obtained via Monte-Carlo simulation of the system model in Fig. 1 by implementing the channel model in (1) are also shown. Moreover, the so-called HD forward error correction (FEC) threshold of 3.84×10^{-3} for the coding rate 0.937 from [13, Table I] is shown. The pre-FEC BER must be lower than this HD FEC threshold in order to achieve a post-FEC BER lower than 10^{-15} . Finally, we define a parameter Δ , which appears in Fig. 2 (both left and right), as the difference in transmit power P at which (16) and the BER approximation, (17) or (18), reach the HD FEC limit.

We first observe from Fig. 2 (right) that Monte-Carlo simulations, as expected, match precisely the exact BER expression (16) for high (pink) and medium (yellow) pointing errors. For low (green) pointing errors, (16) cannot be evaluated analytically as mentioned above, since the corresponding operating point is in the red region in Fig. 2 (left). We then observe that for the (pink) operating point with high pointing errors, the BER approximation (17) is $\Delta = 0.18$ dB away from the exact BER (16) (and hence from the Monte-Carlo simulation result) at the HD FEC limit. For the (yellow) operating point with medium pointing errors, the gap reduces to $\Delta = 0.02$ dB. In this case, the dashed line corresponding to (17) is so close to the exact BER that it is barely visible. Finally, for the (green) operating point with low pointing errors, the gap (now only to the Monte-Carlo simulation results) virtually disappears. Consequently, we conclude that for weak turbulence ($\sigma_R^2 \leq 1$), the proposed average BER approximation (17) is extremely accurate for low pointing errors, while causing a small (< 0.2 dB) inaccuracy for high pointing errors.

From Fig. 2 (right), we also observe that the accuracy of the BER approximation (18) is worse than that of (17) for low (green) and medium (yellow) pointing errors, i.e., $\Delta \approx 0.7$ and 0.4 dB, respectively. Only for high (pink) pointing errors, (18) closely matches (17). We conclude that (18) is mostly suitable for use in the high pointing errors regime while it is a more easily computable approximation.

Lastly, we see in Fig. 2 (right) that for the BER values of interest, i.e., $10^{-2} - 10^{-4}$, operating in high (pink) pointing errors regime the slope of the approximations differs from the correct slope. Only for low (green) and medium (yellow) pointing errors the approximations exhibit the correct slope.

Finally, we revisit Fig. 2 (left) to investigate the accuracy of (17) for other operating points. In addition to the three operating points (green, yellow, pink) studied in Fig. 2 (right), we estimate Δ for six more operating points (color-coded with blue). For all nine settings, Δ value is shown next to the circle markers in Fig. 2 (left) where the size of the circle represents the gap Δ . We verify that the accuracy of the BER approximation (17) is $\Delta \leq 0.18$ dB for all considered pairs of parameters. Moreover, we see that for increasing atmospheric

turbulence, i.e., for increasing σ_R^2 , and/or for decreasing pointing errors, i.e., for decreasing σ_s , the accuracy of the approximation improves and Δ approaches 0 dB.

6. CONCLUSIONS

In this work we developed a closed-form and simple approximation for the PDF that jointly models atmospheric turbulence, pointing errors, and atmospheric losses, under weak turbulence regime. Our proposed expressions provide a more computationally-efficient solution to the ones available in the literature. Additionally, approximate analytical expressions of BER were derived based on our proposed PDF. From the results obtained, we can conclude that for all the applicable regions considered in this work, the approximate BER (17) is in close agreement with the true BER obtained via Monte-Carlo simulations with a maximum error of 0.18 dB, while (18) is a worse approximation with a maximum error of 0.71 dB. Future works will entail the pursuod of alternative approximations to ensure the correct slope of BER results within the high pointing errors regime.

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